Formative Assessment 5

APM1220 - Applied Multivariate Data Analysis

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Introduction

For this assessment, Principal Component Analysis (PCA) will be employed using the dataset of the World Happiness Report 2021 to analyze the underlying dimensions of cross-country well-being. Additionally this analysis will use seven quantitative variables: Ladder Score, Logged GDP per capita, Social Support, Healthy Life Expectancy, Freedom to Make Life Choices, Generosity, and Perceptions of Corruption.

Since each variable are measured on different scales, the variables will be standardized before conducting PCA using the correlation matrix. The analysis will also employ the usages of descriptive statistics, a correlation matrix, eigenvalues, proportions of variance explained, component coefficients, and a scree plot. Afterwards, the appropriate number of principal components will then be determined using the scree plot, the Kaiser criterion, cumulative variance explained, and the interpretability of the components.

Part A – Data Preparation and Descriptive Analysis

Identify the number of observations and the seven variables used in the analysis.

```
happiness <- read.csv(
  "world-happiness-report-2021.csv",
  check.names = FALSE
)

pca_data <- happiness %>%
  select(
    `Ladder score`,
    `Logged GDP per capita`,
    `Social support`,
    `Healthy life expectancy`,
    `Freedom to make life choices`,
    `Generosity`,
    `Perceptions of corruption`
  )

n <- nrow(pca_data)
p <- ncol(pca_data)

cat("Number of observations:", n, "\n")
```

```
## Number of observations: 149
```

```
cat("Number of variables:", p, "\n")
```

```
## Number of variables: 7
```

```
names(pca_data)
```

```
## [1] "Ladder score"                "Logged GDP per capita"
## [3] "Social support"              "Healthy life expectancy"
## [5] "Freedom to make life choices" "Generosity"
## [7] "Perceptions of corruption"
```

Interpretation: The dataset contains 149 observations and 7 quantitative variables, each of the selected variables measures the one of the following: Ladder score, Logged GDP per capita, Social support, Healthy life expectancy, Freedom to make life choices, Generosity, and Perceptions of corruption.

Compute the mean and standard deviation for each variable.

The sample mean of a variable is

$$\bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$$

and its sample standard deviation is

$$s_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}$$

```
descriptive_stats <- data.frame(
  Variable = names(pca_data),
  Mean = sapply(pca_data, mean),
  Standard_Deviation = sapply(pca_data, sd)
)

kable(
  descriptive_stats,
  digits = 3,
  caption = "Descriptive Statistics of the Seven Variables"
)
```

Table 1: Descriptive Statistics of the Seven Variables

	Variable	Mean	Standard_Deviation
Ladder score	Ladder score	5.533	1.074
Logged GDP per capita	Logged GDP per capita	9.432	1.159
Social support	Social support	0.815	0.115
Healthy life expectancy	Healthy life expectancy	64.993	6.762

	Variable	Mean	Standard_Deviation
Freedom to make life choices	Freedom to make life choices	0.792	0.113
Generosity	Generosity	-0.015	0.151
Perceptions of corruption	Perceptions of corruption	0.727	0.179

Interpretation: The descriptive statistics show that the variables have different numerical scales. For example, Healthy Life Expectancy has a mean of 64.993, while Social Support has a mean of 0.815. Generosity has a slightly negative mean of -0.015. These differences in scale support the need to standardize the variables before conducting PCA.

Explain why the variables should be standardized before conducting PCA.

Standardization transforms each variable into a standardized score using

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where x_{ij} is the observed value, $\bar{x}_j$ is the sample mean, and s_j is the sample standard deviation of variable j.

```
# Standardize the seven quantitative variables
Z <- scale(pca_data)

# Display the means and standard deviations after standardization
standardization_check <- data.frame(
  Variable = colnames(Z),
  Mean = colMeans(Z),
  Standard_Deviation = apply(Z, 2, sd)
)

kable(
  standardization_check,
  digits = 3,
  caption = "Verification of Standardized Variables"
)
```

Table 2: Verification of Standardized Variables

	Variable	Mean	Standard_Deviation
Ladder score	Ladder score	0	1
Logged GDP per capita	Logged GDP per capita	0	1
Social support	Social support	0	1
Healthy life expectancy	Healthy life expectancy	0	1
Freedom to make life choices	Freedom to make life choices	0	1
Generosity	Generosity	0	1
Perceptions of corruption	Perceptions of corruption	0	1

Interpretation: After standardization, each variable has a mean approximately equal to zero and a standard deviation equal to one. This ensures that variables measured on larger numerical scales do not disproportionately influence the principal components. Standardization also makes the PCA equivalent to performing an eigen-decomposition of the correlation matrix of the original variables.

Produce and briefly describe the correlation matrix. Identify at least two pairs of variables that show relatively strong positive or negative relationships.

The Pearson correlation coefficient between variables X_j and X_k is

$$r_{jk} = \frac{\text{Cov}(X_j, X_k)}{s_j s_k}.$$

The correlation matrix summarizes the direction and strength of the linear relationships among the seven variables.

```
# Compute the correlation matrix
cor_matrix <- cor(pca_data, method = "pearson")

kable(
  cor_matrix,
  digits = 3,
  caption = "Correlation Matrix of the Seven Well-Being Variables"
)
```

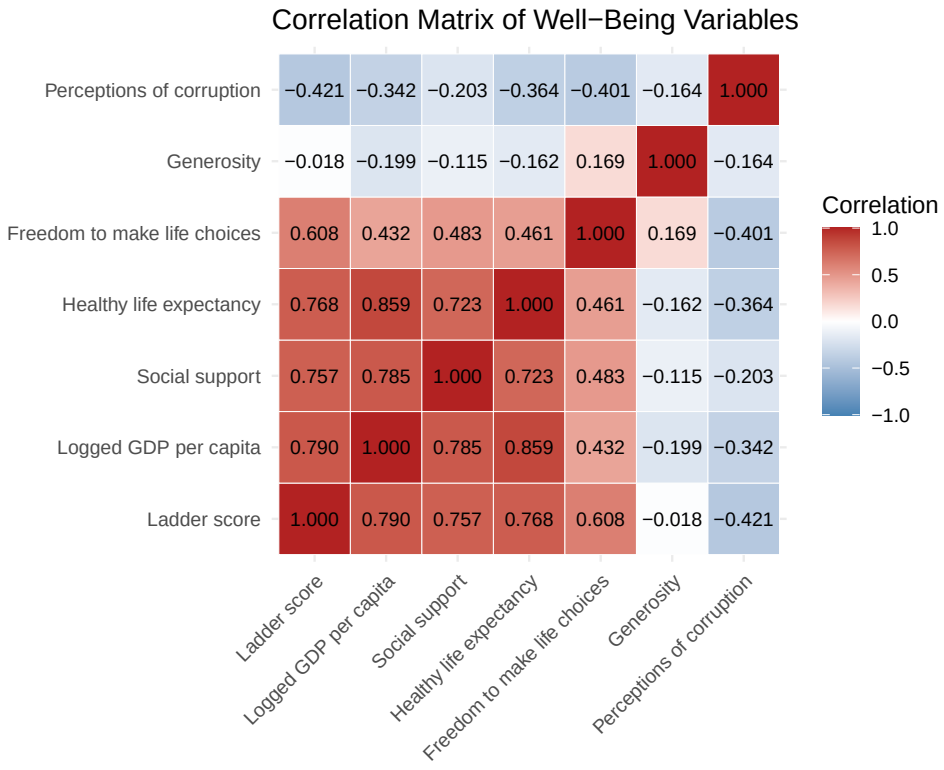
Table 3: Correlation Matrix of the Seven Well-Being Variables

	Ladder score	Logged GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
Ladder score	1.000	0.790	0.757	0.768	0.608	-0.018	-0.421
Logged GDP per capita	0.790	1.000	0.785	0.859	0.432	-0.199	-0.342
Social support	0.757	0.785	1.000	0.723	0.483	-0.115	-0.203
Healthy life expectancy	0.768	0.859	0.723	1.000	0.461	-0.162	-0.364
Freedom to make life choices	0.608	0.432	0.483	0.461	1.000	0.169	-0.401
Generosity	-0.018	-0.199	-0.115	-0.162	0.169	1.000	-0.164
Perceptions of corruption	-0.421	-0.342	-0.203	-0.364	-0.401	-0.164	1.000

```
# Visualize the correlation matrix
cor_long <- as.data.frame(as.table(cor_matrix))

ggplot(
  cor_long,
  aes(x = Var1, y = Var2, fill = Freq)
) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = sprintf("%.3f", Freq)),
    size = 3
  ) +
```

```
  scale_fill_gradient2(
    low = "steelblue",
    mid = "white",
    high = "firebrick",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  labs(
    title = "Correlation Matrix of Well-Being Variables",
    x = NULL,
    y = NULL,
    fill = "Correlation"
  ) +
  coord_equal() +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
```



Interpretation: The correlation matrix reveals several strong positive relationships with Logged GDP per capita and Healthy Life Expectancy having a correlation of 0.837. This suggests that countries with higher logged GDP per capita tend to have higher healthy life expectancy. Ladder Score is also strongly positively correlated with Logged GDP per capita (0.790), Healthy Life Expectancy (0.768), and Social Support (0.757).

Negative relationships are also present, with Ladder Score and Perceptions of Corruption having a correlation of -0.421. Whilst Freedom to Make Life Choices and Perceptions of Corruption have a correlation of -0.401.

Generosity also showed to have relatively weak correlations with most other variables, including Ladder Score (-0.018).

These relationships suggest that several measures of cross-country well-being are associated with one another, supporting the use of PCA to investigate whether the seven variables can be summarized by a smaller number of principal components.

Part B – Principal Component Analysis

Conduct PCA Using the Correlation Matrix

Principal Component Analysis (PCA) is conducted using the seven standardized quantitative variables. The correlation matrix is used so that all variables contribute on a comparable scale, regardless of their original units.

The standardized value of each observation is calculated as:

$$Z_{ij} = \frac{X_{ij} - \bar{X}_j}{s_j}$$

where X_{ij} is the original value, $\bar{X}_j$ is the mean of variable j, and s_j is its sample standard deviation.

PCA is performed by decomposing the correlation matrix R:

$$R\mathbf{v}_k = \lambda_k \mathbf{v}_k$$

where λ_k is the eigenvalue and $\mathbf{v}_k$ is the corresponding eigenvector.

```
standardized_data <- scale(pca_data)

cor_matrix <- cor(standardized_data, use = "complete.obs")

pca_result <- eigen(cor_matrix)

round(cor_matrix, 3)
```

```
##                               Ladder score Logged GDP per capita Social support
## Ladder score                         1.000                 0.790          0.757
## Logged GDP per capita                0.790                 1.000          0.785
## Social support                       0.757                 0.785          1.000
## Healthy life expectancy              0.768                 0.859          0.723
## Freedom to make life choices         0.608                 0.432          0.483
## Generosity                          -0.018                -0.199         -0.115
## Perceptions of corruption           -0.421                -0.342         -0.203
##                               Healthy life expectancy
## Ladder score                                    0.768
## Logged GDP per capita                           0.859
## Social support                                  0.723
## Healthy life expectancy                         1.000
## Freedom to make life choices                    0.461
## Generosity                                     -0.162
## Perceptions of corruption                      -0.364
```

```
##                             Freedom to make life choices Generosity
## Ladder score                                       0.608     -0.018
## Logged GDP per capita                              0.432     -0.199
## Social support                                     0.483     -0.115
## Healthy life expectancy                            0.461     -0.162
## Freedom to make life choices                       1.000      0.169
## Generosity                                         0.169      1.000
## Perceptions of corruption                         -0.401     -0.164
##                             Perceptions of corruption
## Ladder score                                   -0.421
## Logged GDP per capita                          -0.342
## Social support                                 -0.203
## Healthy life expectancy                        -0.364
## Freedom to make life choices                   -0.401
## Generosity                                     -0.164
## Perceptions of corruption                       1.000
```

Report the Eigenvalues of the Seven Principal Components

The eigenvalues measure the amount of total variance captured by each principal component. Because seven standardized variables are analyzed, the total variance is seven.

$$\sum_{k=1}^{7} \lambda_k = 7$$

```
eigenvalues <- pca_result$values

eigenvalue_table <- data.frame(
  Component = paste0("PC", 1:7),
  Eigenvalue = eigenvalues
)

kable(
  transform(eigenvalue_table,
            Eigenvalue = round(Eigenvalue, 3)),
  caption = "Eigenvalues of the Seven Principal Components"
)
```

Table 4: Eigenvalues of the Seven Principal Components

Component	Eigenvalue
PC1	3.914
PC2	1.289
PC3	0.708
PC4	0.518
PC5	0.251
PC6	0.193
PC7	0.126

Interpretation: PC1 has the largest eigenvalue, followed by PC2, the remaining components have progressively smaller eigenvalues, suggesting that the first components encapsulates more of the total variation in the data than the later components.

Report the Proportion of Variance Explained by Each Component

The proportion of variance explained by component k is calculated by dividing its eigenvalue by the total variance:

$$\text{Proportion of Variance}_k = \frac{\lambda_k}{\sum_{j=1}^{7} \lambda_j} = \frac{\lambda_k}{7}$$

```
proportion_variance <- eigenvalues / sum(eigenvalues)

proportion_table <- data.frame(
  Component = paste0("PC", 1:7),
  Eigenvalue = eigenvalues,
  Proportion_Variance = proportion_variance
)

kable(
  transform(
    proportion_table,
    Eigenvalue = round(Eigenvalue, 3),
    Proportion_Variance = round(Proportion_Variance, 3)
  ),
  caption = "Proportion of Variance Explained by Each Component"
)
```

Table 5: Proportion of Variance Explained by Each Component

Component	Eigenvalue	Proportion_Variance
PC1	3.914	0.559
PC2	1.289	0.184
PC3	0.708	0.101
PC4	0.518	0.074
PC5	0.251	0.036
PC6	0.193	0.028
PC7	0.126	0.018

Interpretation: PC1 explains approximately 55.9% of the total variance, while PC2 explains approximately 18.4%. Each subsequent component decreases accounting for a smaller proportion of the total variance.

Report the Cumulative Proportion of Variance Explained

The cumulative proportion of variance explained is the sum of the individual proportions up to a given component:

$$\text{Cumulative Proportion}_k = \sum_{j=1}^{k} \frac{\lambda_j}{7}$$

```r
cumulative_variance <- cumsum(proportion_variance)

cumulative_table <- data.frame(
  Component = paste0("PC", 1:7),
  Proportion_Variance = proportion_variance,
  Cumulative_Variance = cumulative_variance
)

kable(
  transform(
    cumulative_table,
    Proportion_Variance = round(Proportion_Variance, 3),
    Cumulative_Variance = round(Cumulative_Variance, 3)
  ),
  caption = "Individual and Cumulative Proportion of Variance Explained"
)
```

Table 6: Individual and Cumulative Proportion of Variance Explained

Component	Proportion_Variance	Cumulative_Variance
PC1	0.559	0.559
PC2	0.184	0.743
PC3	0.101	0.844
PC4	0.074	0.918
PC5	0.036	0.954
PC6	0.028	0.982
PC7	0.018	1.000

Interpretation: The first two principal components jointly explain approximately 74.3% of the total variance, including PC3 raises the cumulative proportion to approximately 84.4%. Thus, two components provide a substantial summary of the original seven variables, while three components retain more of the total variation.

Present the Eigenvectors/Component Coefficients for the Retained Components

The eigenvectors provide the coefficients used to form the principal components from the standardized variables. For the first two components:

$$PC_k = v_{1k}Z_1 + v_{2k}Z_2 + \cdots + v_{7k}Z_7$$

For this section, the first two components are presented because their eigenvalues exceed 1 and together explain approximately 74.3% of the total variance.

```r
eigenvectors <- pca_result$vectors[, 1:2]

if (eigenvectors[1, 1] < 0) {
  eigenvectors[, 1] <- -eigenvectors[, 1]
}

if (eigenvectors[6, 2] < 0) {
```

```
  eigenvectors[, 2] <- -eigenvectors[, 2]
}

coefficient_table <- data.frame(
  Variable = rownames(cor_matrix),
  PC1 = eigenvectors[, 1],
  PC2 = eigenvectors[, 2],
  row.names = NULL
)

kable(
  transform(
    coefficient_table,
    PC1 = round(PC1, 3),
    PC2 = round(PC2, 3)
  ),
  caption = "Eigenvectors / Component Coefficients for PC1 and PC2"
)
```

Table 7: Eigenvectors / Component Coefficients for PC1 and PC2

Variable	PC1	PC2
Ladder score	0.464	0.040
Logged GDP per capita	0.459	-0.197
Social support	0.433	-0.163
Healthy life expectancy	0.453	-0.146
Freedom to make life choices	0.341	0.365
Generosity	-0.042	0.753
Perceptions of corruption	-0.252	-0.460

Interpretation: PC1 has relatively large positive coefficients for Ladder score, Logged GDP per capita, Healthy life expectancy, and Social support. While PC2 has its largest positive coefficient for Generosity and a negative coefficient for Perceptions of corruption. These coefficients suggest how each standardized variable contributes to the respective component; their signs describe the direction of contribution, not causation.

Identify the Variables with the Largest Absolute Coefficients in the First Principal Component

The absolute values of the PC1 coefficients indicate the relative magnitude of each variable's contribution to the first component. The variables are ranked using:

$$|v_{j1}|$$

where v_{j1} is the coefficient of variable j in PC1.

```
pc1_importance <- data.frame(
  Variable = rownames(cor_matrix),
  PC1_Coefficient = eigenvectors[, 1],
  Absolute_Coefficient = abs(eigenvectors[, 1])
) %>%
```

```
  arrange(desc(Absolute_Coefficient))

kable(
  transform(
    pc1_importance,
    PC1_Coefficient = round(PC1_Coefficient, 3),
    Absolute_Coefficient = round(Absolute_Coefficient, 3)
  ),
  caption = "Variables Ranked by Absolute PC1 Coefficients"
)
```

Table 8: Variables Ranked by Absolute PC1 Coefficients

Variable	PC1_Coefficient	Absolute_Coefficient
Ladder score	0.464	0.464
Logged GDP per capita	0.459	0.459
Healthy life expectancy	0.453	0.453
Social support	0.433	0.433
Freedom to make life choices	0.341	0.341
Perceptions of corruption	-0.252	0.252
Generosity	-0.042	0.042

Interpretation: The variables with the largest absolute coefficients in PC1 are Ladder score, Logged GDP per capita, Healthy life expectancy, and Social support. These variables contribute most strongly to the first principal component. Together with the positive coefficient for Freedom to make life choices, their pattern supports interpreting PC1 as a general well-being and socioeconomic advantage dimension.

Part C – Determining the Number of Components

Construct a Scree Plot of the Eigenvalues

A scree plot displays the eigenvalues of the principal components in descending order. It helps determine how much variance each component captures and where the contribution of additional components begins to decrease substantially.

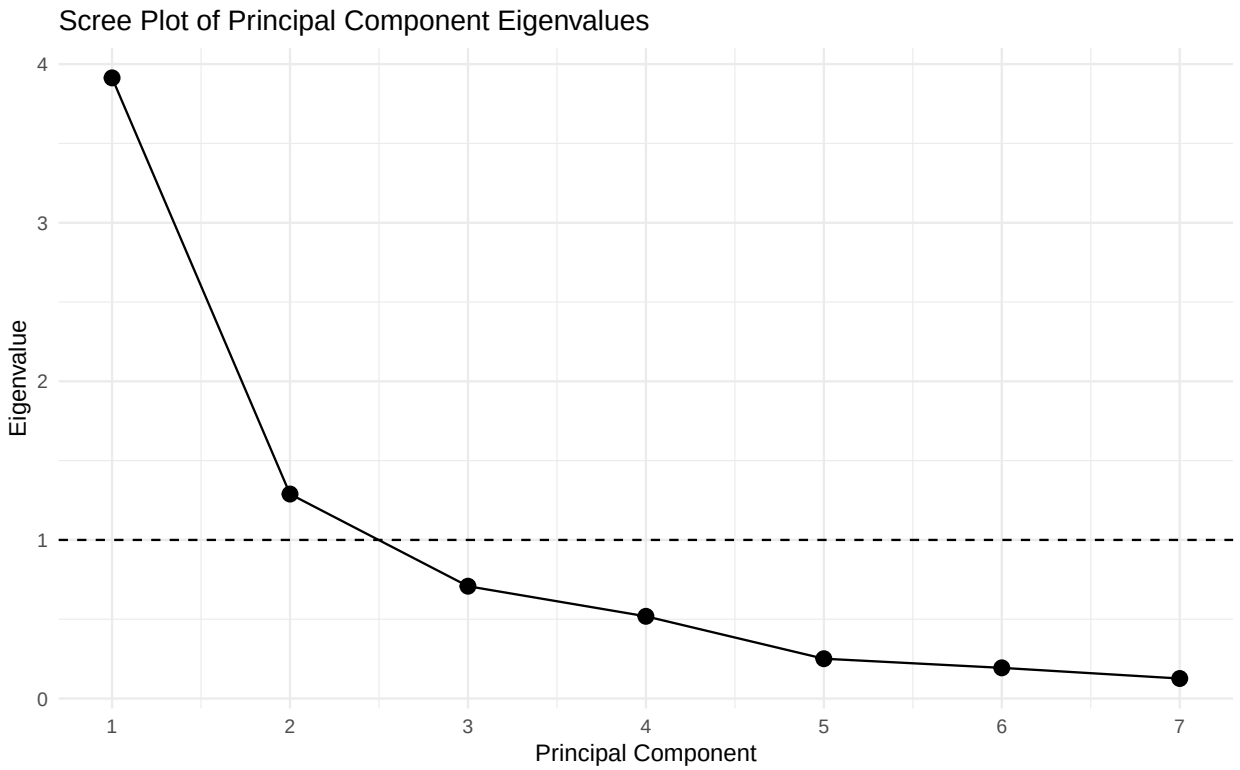
The eigenvalues are ordered such that:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_7$$

```
scree_data <- data.frame(
  Component = 1:7,
  Eigenvalue = eigenvalues
)

ggplot(scree_data, aes(x = Component, y = Eigenvalue)) +
  geom_line() +
  geom_point(size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  scale_x_continuous(breaks = 1:7) +
  labs(
```

```
    title = "Scree Plot of Principal Component Eigenvalues",
    x = "Principal Component",
    y = "Eigenvalue"
  ) +
  theme_minimal()
```



Interpretation: The scree plot shows that PC1 has the largest eigenvalue, followed by a substantial decrease at PC2. The eigenvalues decline more gradually after PC2, suggesting that the first two components capture the most prominent patterns of variation in the data.

Identify the Apparent Elbow in the Scree Plot (2 points)

The elbow is the point where the eigenvalue curve changes from a relatively steep decline to a more gradual decrease.

```
kable(
  data.frame(
    Component = paste0("PC", 1:7),
    Eigenvalue = round(eigenvalues, 3)
  ),
  caption = "Eigenvalues Used to Identify the Scree Plot Elbow"
)
```

Table 9: Eigenvalues Used to Identify the Scree Plot Elbow

Component	Eigenvalue
PC1	3.914
PC2	1.289
PC3	0.708
PC4	0.518
PC5	0.251
PC6	0.193
PC7	0.126

Interpretation: The apparent elbow occurs around PC2 to PC3 because the eigenvalues decrease sharply from PC1 to PC2 and then fall below 1 at PC3, after which the decline is becomes more apparent. This suggests that retaining the first two principal components seems to be the most appropriate approach.

Apply the Kaiser Criterion

The Kaiser criterion retains components with eigenvalues greater than 1:

$$\lambda_k > 1$$

This criterion is commonly applied to PCA based on a correlation matrix because each standardized variable contributes one unit of variance.

```
kaiser_table <- data.frame(
  Component = paste0("PC", 1:7),
  Eigenvalue = eigenvalues,
  Retain = eigenvalues > 1
)

kable(
  transform(
    kaiser_table,
    Eigenvalue = round(Eigenvalue, 3)
  ),
  caption = "Kaiser Criterion for Component Retention"
)
```

Table 10: Kaiser Criterion for Component Retention

Component	Eigenvalue	Retain
PC1	3.914	TRUE
PC2	1.289	TRUE
PC3	0.708	FALSE
PC4	0.518	FALSE
PC5	0.251	FALSE
PC6	0.193	FALSE
PC7	0.126	FALSE

```r
retained_kaiser <- which(eigenvalues > 1)
paste("Components retained:", paste0("PC", retained_kaiser, collapse = ", "))
```

```
## [1] "Components retained: PC1, PC2"
```

Determine an Appropriate Number of Components to Retain (2 points)

The decision considers the scree plot, the Kaiser criterion, and the cumulative proportion of variance explained.

The cumulative proportion explained by the first k components is:

$$C_k = \sum_{j=1}^{k} \frac{\lambda_j}{7}$$

```r
retention_table <- data.frame(
  Component = paste0("PC", 1:7),
  Eigenvalue = eigenvalues,
  Proportion_Variance = proportion_variance,
  Cumulative_Variance = cumulative_variance
)

kable(
  transform(
    retention_table,
    Eigenvalue = round(Eigenvalue, 3),
    Proportion_Variance = round(Proportion_Variance, 3),
    Cumulative_Variance = round(Cumulative_Variance, 3)
  ),
  caption = "Summary for Selecting the Number of Principal Components"
)
```

Table 11: Summary for Selecting the Number of Principal Components

Component	Eigenvalue	Proportion_Variance	Cumulative_Variance
PC1	3.914	0.559	0.559
PC2	1.289	0.184	0.743
PC3	0.708	0.101	0.844
PC4	0.518	0.074	0.918
PC5	0.251	0.036	0.954
PC6	0.193	0.028	0.982
PC7	0.126	0.018	1.000

Interpretation: From the results above, two principal components seems to be the most appropriate number of components to retain. This decision is supported by the apparent elbow around PC2–PC3 and the Kaiser criterion, which retains PC1 and PC2. Together, the first two components explain approximately 74.3% of the total variance, providing a substantial summary of the original seven variables. Retaining a third component would increase the cumulative variance explained to approximately 84.4%, but PC3 has an eigenvalue below 1 and contributes less variance than an individual standardized variable.

Explain Why Multiple Criteria Should Be Used

The number of components should not be determined using only one criterion because each method has limitations. The scree plot relies partly on visual judgment, the Kaiser criterion uses a fixed eigenvalue threshold, and cumulative variance depends on how much information the researcher wants to preserve. Considering all three provides a more balanced decision and reduces the risk of retaining too many or too few components.

Part D – Interpretation of Principal Components

Interpret Principal Component 1 (PC1)

Principal Component 1 is a linear combination of the seven standardized variables:

$$PC_1 = \sum_{j=1}^{7} v_{j1} Z_j$$

where v_{j1} represents the coefficient of variable j in PC1 and Z_j represents its standardized value.

```
# Display PC1 coefficients in descending order of absolute magnitude
pc1_interpretation <- data.frame(
  Variable = rownames(cor_matrix),
  Coefficient = eigenvectors[, 1]
) %>%
  arrange(desc(abs(Coefficient)))

kable(
  transform(
    pc1_interpretation,
    Coefficient = round(Coefficient, 3)
  ),
  caption = "PC1 Coefficients Ranked by Absolute Magnitude"
)
```

Table 12: PC1 Coefficients Ranked by Absolute Magnitude

Variable	Coefficient
Ladder score	0.464
Logged GDP per capita	0.459
Healthy life expectancy	0.453
Social support	0.433
Freedom to make life choices	0.341
Perceptions of corruption	-0.252
Generosity	-0.042

Interpretation: PC1 has its largest positive coefficients for Ladder score (0.464), Logged GDP per capita (0.459), Healthy life expectancy (0.453), and Social support (0.433). Freedom to make life choices also has a positive coefficient (0.341). Perceptions of corruption has a negative coefficient (-0.252), while Generosity has a coefficient close to zero (-0.042). Therefore, PC1 primarily represents a dimension of cross-country

well-being associated with higher reported life evaluations, greater economic prosperity, longer healthy life expectancy, stronger social support, and greater freedom. The negative coefficient for Perceptions of corruption indicates an opposing direction along this component, while Generosity contributes very little to PC1.

Interpret Principal Component 2 (PC2)

Principal Component 2 is another linear combination of the standardized variables:

$$PC_2 = \sum_{j=1}^{7} v_{j2} Z_j$$

```
# Display PC2 coefficients in descending order of absolute magnitude
pc2_interpretation <- data.frame(
  Variable = rownames(cor_matrix),
  Coefficient = eigenvectors[, 2]
) %>%
  arrange(desc(abs(Coefficient)))

kable(
  transform(
    pc2_interpretation,
    Coefficient = round(Coefficient, 3)
  ),
  caption = "PC2 Coefficients Ranked by Absolute Magnitude"
)
```

Table 13: PC2 Coefficients Ranked by Absolute Magnitude

Variable	Coefficient
Generosity	0.753
Perceptions of corruption	-0.460
Freedom to make life choices	0.365
Logged GDP per capita	-0.197
Social support	-0.163
Healthy life expectancy	-0.146
Ladder score	0.040

Interpretation: PC2 is dominated by Generosity (0.753), followed by Perceptions of corruption (-0.460) and Freedom to make life choices (0.365). Logged GDP per capita (-0.197), Social support (-0.163), and Healthy life expectancy (-0.146) have smaller negative coefficients, while Ladder score (0.040) contributes very little. Thus, PC2 mainly contrasts higher Generosity and Freedom to make life choices with higher values of the Perceptions of corruption variable, while the remaining variables contribute less strongly. This component describes a different pattern of variation from PC1 and should not be interpreted as evidence of a causal relationship.

Give a Descriptive Name or Label for PC1 and Explain the Choice

Proposed label: General Well-Being and Socioeconomic Advantage

PC1 is labeled General Well-Being and Socioeconomic Advantage because its largest positive coefficients correspond to Ladder score, Logged GDP per capita, Healthy life expectancy, and Social support with Freedom to make life choices also contributing positively. These variables collectively represent life evaluation, economic conditions, health, social support, and personal freedom. The label summarizes the dominant pattern in PC1 without suggesting that any one variable causes another.

Explain How Many Original Variables Can Potentially Be Represented by the Retained Components

The analysis retains two principal components from the original seven variables. Each component is a weighted combination of all seven standardized variables, although some variables contribute more strongly than others.

$$\text{Original variables} = 7$$

$$\text{Retained components} = 2$$

Interpretation: The two retained components (PC1 & PC2) summarize the information in all seven original variables using two new dimensions. Therefore, the seven original variables are represented by two components, achieving dimensionality reduction from seven variables to two while preserving approximately 74.3% of the total variance. Noting this, the two components still does not reproduce every original value exactly; with approximately 25.7% of the total variance not captured by the first two components.

Underlying Dimensions of Cross-Country Well-Being

Conclusion: PCA reveals that cross-country well-being in the 2021 World Happiness Report can be summarized by two principal dimensions. PC1 represents general well-being and socioeconomic advantage, with strong positive contributions from life evaluation, GDP per capita, healthy life expectancy, social support, and freedom. PC2 captures a contrasting pattern involving generosity, freedom, and perceptions of corruption that is largely distinct from the dominant PC1 pattern. Together, these two components explain approximately 74.3% of the total variance among the seven variables. This indicates that a substantial amount of the variation in cross-country well-being can be summarized in two dimensions, while the remaining variation is distributed across the other five components.

Overall Conclusion

Concluding this report, the Principal Component Analysis (PCA) of the 2021 World Happiness Report examined seven quantitative variables across 149 countries to identify the several underlying dimensions of cross-country well-being. The correlation analysis revealed several notable relationships among life evaluation, GDP per capita, social support, and healthy life expectancy. Using the correlation matrix of standardized variables, PCA identified two principal components (PC1 & PC2) that covered approximately 74.3% of the total variance. Overall, the results demonstrate that cross-country well-being can be summarized using two dimensions, simplifying the original seven variables while preserving a substantial amount of information.